Masina de inductie trifazata cu rotor in colivie
2.Elemente de proiectare a masinii de inductie

Date de intrare:

$m_1 := 3$ Conexiune stea(Y)

$P_N := 55\ kW$ Clasa de izolatie: F(max 120°C)

$U_N := 400\ V$ Grupa de protectie: IP22

$f_1 := 50\ Hz$ Serviciul : S1 (Continuu)

$n_N := 1480\ min^{-1}$ Conditii de funcitionare: Normale

2.1 Determinarea principalelor mărimi de calcul ale mașinii de inducție

2.1.1 Numarul de perechi de poli

$p := 2$ $n_1 := 1500\ min^{-1}$

2.1.2 Puterea aparenta nominala absorbita si curentul nominal de faza

$U_1 := \dfrac{U_N}{\sqrt{3}} = 230.94\ V$ $\varphi := \mathrm{acos}(0.895)$

$\eta_N := 0.91$

$S_N := \dfrac{P_N}{\eta_N \cdot \cos(\varphi)} = 67.53\ kV \cdot A$ $S_N = 67530.235\ V \cdot A$

$I_{1N} := \dfrac{S_N}{3 \cdot U_1} = 97.471\ A$ $I_{Nl} := I_{1N} = 97.471\ A$

2.1.3 Tensiunea electromotoare de faza si puterea aparenta interioara

$k_E := 0.98 - 0.005 \cdot p = 0.97$

$E_1 := k_E \cdot U_1 = 224.012\ V$

$S_{iN} := 3 \cdot E_1 \cdot I_{1N} = 65.504\ kV \cdot A$ $\boxed{S_{iN}} := k_E \cdot S_N = 65504.328\ V \cdot A$

$k_{w1} := 0.92$

$k_{sd} := 1.3$

$k_B := 1.09$

$\alpha_i := 0.7$

2.1.4 Diametrul interior al statorului

$\lambda := 0.8$ $\qquad$ $dm := 100\ mm$ transformare de unitati de masura

$$C := 200 \cdot \frac{J}{dm^3}$$

$$D_1 := 23.33 \sqrt[3]{\frac{p^2 \cdot s}{\lambda} \cdot \frac{S_{iN}}{C \cdot 1000000}} = 274.991\ mm \qquad D := 275\ mm$$

2.1.5 Diametrul exterior al statorului

$k_D := 1.48$

$$D_e := k_D \cdot D = 407\ mm \qquad D_e := 410\ mm$$

2.1.6 Lungimea geometrica a masinii

$$A := 390\ \frac{A}{cm}$$

$B_\delta := 0.75\ T$

$$l_g := \frac{2 \cdot p \cdot S_{iN} \cdot s}{k_B \cdot k_{w1} \cdot \alpha_i \cdot \pi^2 \cdot D^2 \cdot A \cdot B_\delta \cdot 100} = 170.973\ mm$$

$$l_g := 171\ mm = 17.1\ cm$$

$$\tau := \frac{\pi \cdot D}{2 \cdot p} = 21.598\ cm \qquad \tau := 21.6 \cdot cm$$

Verificare prin pas polar:

$$\lambda := \frac{l_g}{\tau} = 0.792$$

2.1.7 Lungimea intrefierului

$$\delta := 0.03 \cdot \left(4\ mm + 0.07 \cdot \sqrt{D \cdot l_g}\right) = 0.575\ mm \qquad \delta := 0.55\ mm$$

3.ÎNFĂȘURĂRILE ȘI CRESTĂTURILE STATORULUI

3.1 Numarul de crestaturi ale statorului

$q_1 := 6$

$Z_1 := 2 \cdot p \cdot m_1 \cdot q_1 = 72$

$t_1 := \frac{\pi \cdot D}{Z_1} = 11.999\ mm$ $\quad\quad$ $\boxed{t_1} := 12 \cdot mm$

3.2 Elementele infasurarii statorului

$\phi := \alpha_i \cdot \tau \cdot l_g \cdot B_\delta = 0.019\ Wb$

$a_1 := 2$

$w_1 := \frac{k_E \cdot U_1}{4 \cdot k_B \cdot f_1 \cdot k_{w1} \cdot \phi} = 57.599$ $\quad\quad$ $\boxed{w_1} := 59$

$n_{c1} := \frac{2 \cdot m_1 \cdot a_1 \cdot w_1}{Z_1} = 9.833$ $\quad\quad$ $\boxed{n_{c1}} := 10$

$\boxed{w_1} := \frac{Z_1 \cdot n_{c1}}{2 \cdot 3 \cdot a_1} = 60$

$\boxed{A} := \frac{Z_1 \cdot n_{c1} \cdot I_{1N}}{\pi \cdot D \cdot a_1} = 406.16\ \frac{A}{cm}$ $\quad\quad$ $\varepsilon_A := 3.97\%$

$\boxed{\phi} := \frac{k_E \cdot U_1}{4 \cdot k_B \cdot f_1 \cdot k_{w1} \cdot w_1} = 0.019\ Wb$

$\boxed{B_\delta} := \frac{\phi}{\alpha_i \cdot \tau \cdot l_g} = 0.72\ T$ $\quad\quad$ $\varepsilon_{B\delta} := 4.16\%$

3.3 Tipul Infasurarii si dimensiunile crestaturii statorului

$J_1 := 5.5\ \frac{A}{mm^2}$

$S_{Cu1} := \frac{I_{1N}}{a_1 \cdot J_1} = 8.861\ mm^2$

$n_f := 4$

$d_c := \sqrt{\frac{4 \cdot S_{Cu1}}{n_f \cdot \pi}} = 1.679\ mm$ $\quad\quad$ $\boxed{d_c} := 1.7\ mm$

$n_{tot} := n_f \cdot n_{c1} = 40$

$s_{cond} := \frac{\pi \cdot d_c^{\ 2}}{4} = 2.27\ mm^2$

$\boxed{S_{Cu1}} := n_f \cdot s_{cond} = 9.079\ mm^2$

$\boxed{J_1} := \dfrac{I_{1N}}{a_1 \cdot S_{Cu1}} = 5.368\ \dfrac{A}{mm^2}$

$iz := \dfrac{d_{ci} - d_c}{2} = 0.043\ mm$

$b_{d1} := \dfrac{t_1 \cdot B_\delta}{k_{Fe} \cdot B_{d1}} = 5.35\ mm$

$S'_{cr1} := \dfrac{n_{tot} \cdot d_{ci}^{\ 2}}{k_u} = 169.932\ mm^2$ $\boxed{S'_{cr1}} := 170\ mm^2$

$d_{ci} := 1.785\ mm$

$B_{d1} := 1.7\ T$

$k_{Fe} := 0.95$

$k_u := 0.75$ $\boxed{iz} := 0.05 \cdot mm$

3.4 Dimensionarea analitica a crestaturii statorice trapezoidale

$b_{istm1} := d_{ci} + 1.5\ mm = 3.285\ mm$ $\boxed{b_{istm1}} := 3.3 \cdot mm$

$h_{istm1} := 1.5\ mm$

$h_{pana} := 3\ mm$

$g_{iz} := 0.3\ mm$

$b_{cr1v} := \dfrac{\pi}{Z_1} \cdot (D + 2 \cdot h_{istm1} + 2 \cdot h_{pana} + 4 \cdot g_{iz}) - b_{d1} = 7.094\ mm$ $\boxed{b_{cr1v}} := 7.1 \cdot mm$

$$h_{util.cr1} := \frac{\sqrt{(b_{cr1v} - 2 \cdot g_{iz})^2 + 4 \cdot S'_{cr1} \cdot \tan\left(\dfrac{\pi}{Z_1}\right)} - (b_{cr1v} - 2 \cdot g_{iz})}{2 \cdot \tan\left(\dfrac{\pi}{Z_1}\right)} = 22.694\ mm$$

$\boxed{h_{util.cr1}} := 22.7 \cdot mm$

$h_{d1} := h_{util.cr1} + h_{istm1} + h_{pana} + 4 \cdot g_{iz} = 28.4\ mm$

$b_{cr1b} := \dfrac{\pi}{Z_1} \cdot (D + 2 \cdot h_{d1}) - b_{d1} = 9.128\ mm$ $\boxed{b_{cr1b}} := 9.15 \cdot mm$

$b_{d1v} := \dfrac{\pi}{Z_1} \cdot (D + 2 \cdot h_{istm1} + 2 \cdot h_{pana}) - b_{cr1v} = 5.292\ mm$ $\boxed{b_{d1v}} := 5.3 \cdot mm$

$b_{d1b} := \dfrac{\pi}{Z_1} \cdot (D + 2 \cdot h_{d1}) - b_{cr1b} = 5.328\ mm$ $\boxed{b_{d1b}} := 5.35 \cdot mm$

$$S_{cr1} := \frac{(b_{cr1v} - 2 \cdot g_{iz}) + (b_{cr1b} - 2 \cdot g_{iz})}{2} \cdot (h_{d1} - h_{istm1} - h_{pana} - 2 \cdot g_{iz}) = 175.333\ mm^2$$

Verificari necesare: $h_{j1} := \dfrac{D_e - D}{2} - h_{d1} = 39.1\ mm$

$B_{j1} := \dfrac{\phi}{2 \cdot k_{Fe} \cdot l_g \cdot h_{j1}} = 1.465\ T$

4.ÎNFĂȘURĂRILE ȘI CRESTĂTURILE ROTORULUI

$J_{2b} := 4 \ \frac{A}{mm^2}$

$J_{2i} := 0.7 \cdot J_{2b} = 2.8 \ \frac{A}{mm^2}$

$\alpha := \frac{2 \cdot \pi \cdot p}{Z_2} = 9.474 \ deg$

$k_i := \frac{\sin\left(\frac{\pi \cdot p}{Z_1}\right)}{\frac{\pi \cdot p}{Z_1}} = 0.999$

$D_r := D - 2 \cdot \delta = 273.9 \ mm$

$t_2 := \frac{\pi \cdot D_r}{Z_2} = 11.322 \ mm$

$Z_2 := 76$ (Nr. crestaturi inclinate)

$m_2 := Z_2 = 76$

$w_2 := 0.5$

$k_I := 0.925$

$k_{w2} := 0.998$

4.1 Dimensionarea crestaturii rotorului

$U_{20} := \frac{k_i \cdot E_1}{2 \cdot w_1 \cdot k_{w1}} = 2.027 \ V$

$I_2 := k_I \cdot \frac{m_1 \cdot w_1 \cdot k_{w1}}{m_2 \cdot w_2 \cdot k_{w2}} \cdot I_{1N} = 393.7 \ A$

$I_b := k_I \cdot \frac{2 \cdot m_1 \cdot w_1 \cdot k_{w1}}{Z_2 \cdot k_i} \cdot I_{1N} = 393.412 \ A$

$I_i := \frac{I_b}{2 \cdot \sin\left(\frac{\pi \cdot p}{Z_2}\right)} = 2382.024 \ A$

$S_b := \frac{I_b}{J_{2b}} = 98.353 \ mm^2$ $\qquad$ $\boxed{S_b} := 98.35 \ mm^2$

$S_i := \frac{I_i}{J_{2i}} = 850.723 \ mm^2$

4.2 Dimensionarea analitica a crestaturii rotorice trapezoidale

$b_{istm2} := 1.5 \ mm$

$h_{istm2} := 1 \ mm$

$B_{d2} := 1.7 \ T$

$$b_{d2} := \frac{t_2 \cdot B_\delta}{k_{Fe} \cdot B_{d2}} = 5.048\ mm \qquad \boxed{b_{d2}} := 5.05 \cdot mm$$

$$b_{cr2v} := \frac{\pi}{Z_2} \cdot \left(D_r - 2 \cdot h_{istm2}\right) - b_{d2} = 6.189\ mm \qquad \boxed{b_{cr2v}} := 6.2 \cdot mm$$

$$h_{cr2} := \frac{Z_2}{2 \cdot \pi} \cdot \left(b_{cr2v} - \sqrt{b_{cr2v}^{\ 2} - \frac{4 \cdot \pi}{Z_2} \cdot S_b}\right) + h_{istm2} = 19.03\ mm \qquad \boxed{h_{cr2}} := 19.05 \cdot mm$$

$$h_{d2} := h_{cr2} = 19.05\ mm$$

$$b_{cr2b} := b_{cr2v} - \frac{2\ \pi}{Z_2} \cdot \left(h_{cr2} - h_{istm2}\right) = 4.708\ mm \qquad \boxed{b_{cr2b}} := 4.7 \cdot mm$$

$$b_{d2v} := \frac{\pi}{Z_2} \cdot \left(D_r - 2 \cdot h_{istm2}\right) - b_{cr2v} = 5.039\ mm \qquad \boxed{b_{d2v}} := 5.05 \cdot mm$$

$$b_{d2b} := \frac{\pi}{Z_2} \cdot \left(D_r - 2 \cdot h_{d2}\right) - b_{cr2b} = 5.047\ mm \qquad \boxed{b_{d2b}} := 5.05 \cdot mm$$

$$\boxed{S_b} := \frac{\left(b_{cr2v} + b_{cr2b}\right)}{2} \cdot \left(h_{cr2} - h_{istm2}\right) = 98.373\ mm^2$$

Verificari necesare:

$$B_{j2} := 1.3\ T$$

$$h'_{j2} := \frac{\phi}{2 \cdot k_{Fe} \cdot l_g \cdot B_{j2}} = 44.074\ mm$$

$$D_{ir} := D_r - 2 \cdot \left(h_{d2} + h'_{j2}\right) = 147.652\ mm$$

$$d_{cap\_ax} := 39\ mm \qquad \boxed{D_{ir}} := 150\ mm$$

$$d_{ax} := d_{cap\_ax} + 10\ mm = 49\ mm$$

$$h_{j2} := \frac{D_r - D_{ir}}{2} - h_{d2} = 42.9\ mm$$

$$\boxed{B_{j2}} := \frac{\phi}{2 \cdot k_{Fe} \cdot l_g \cdot h_{j2}} = 1.336\ T$$

5. CALCULUL CURENTULUI DE MAGNETIZARE

$$\gamma_1 := \frac{\left(\frac{b_{istm1}}{\delta}\right)^2}{5+\frac{b_{istm1}}{\delta}} = 3.273$$

$$\gamma_2 := \frac{\left(\frac{b_{istm2}}{\delta}\right)^2}{5+\frac{b_{istm2}}{\delta}} = 0.963$$

$$\mu_0 := 4\ \pi \cdot 10^{-7}\ \frac{H}{m}$$

$$k_{C1} := \frac{t_1}{t_1-\delta \cdot \gamma_1} = 1.176$$

$$k_{C2} := \frac{t_2}{t_2-\delta \cdot \gamma_1} = 1.189$$

$$k_C := k_{C1} \cdot k_{C2} = 1.399$$

$$B_{d1} = 1.7\ T$$

$$B_{d2} = 1.7\ T$$

$$B_{j1} = 1.465\ T$$

$$B_{j2} = 1.336\ T$$

$$H_{d1} := 50\ \frac{A}{cm}$$

$$H_{d2} := 50\ \frac{A}{cm}$$

$$H_{j1} := 10\ \frac{A}{cm}$$

$$H_{j2} := 3\ \frac{A}{cm}$$

$$\xi_{j1} := 0.35$$

$$\xi_{j2} := 0.4$$

$$U_{m\delta} := 2 \cdot \frac{B_\delta}{\mu_0} \cdot k_C \cdot \delta = 881.628\ A$$

$$U_{md1} := 2 \cdot H_{d1} \cdot h_{d1} = 284\ A$$

$$U_{md2} := 2 \cdot H_{d2} \cdot h_{d2} = 190.5\ A$$

$$k_{sd} := \frac{U_{m\delta}+U_{md1}+U_{md2}}{U_{m\delta}} = 1.538$$

$$L_{j1} := \frac{\pi \cdot (D_e - h_{j1})}{2 \cdot p} = 291.304\ mm$$

$$L_{j2} := \frac{\pi \cdot (D_{ir} + h_{j2})}{2 \cdot p} = 151.503\ mm$$

$$U_{mj1} := \xi_{j1} \cdot H_{j1} \cdot L_{j1} = 101.956\ A$$

$$U_{mj2} := \xi_{j2} \cdot H_{j2} \cdot L_{j2} = 18.18\ A$$

$$U_{mcirc} := U_{m\delta} + U_{md1} + U_{md2} + U_{mj1} + U_{mj2} = 1476.265\ A$$

$$k_S := \frac{U_{mcirc}}{U_{m\delta}} = 1.674$$

$$I_\mu := \frac{p \cdot U_{mcirc}}{0.9 \cdot m_1 \cdot w_1 \cdot k_{w1}} = 19.81\ A$$

$$I_{\mu'} := \frac{I_\mu}{I_{1N}} \cdot 100 = 20.324 \qquad I_{\mu'} := 20.3\%$$

$$I_\mu = 19.81\ A$$

6. CALCULUL PARAMETRILOR ÎNFĂȘURĂRILOR MAȘINII DE INDUCȚIE

$$\rho_{115} := 0.02464\ \Omega \cdot \frac{mm^2}{m}$$

$$\rho_{20} := 0.01784\ \Omega \cdot \frac{mm^2}{m}$$

$$t_{med} := \frac{\pi \cdot (D + h_{cr1})}{Z_1} = 13.238\ mm$$

$$R_m := \frac{y_1 \cdot t_{med}}{2} = 99.287\ mm$$

$$l_{f1} := \pi \cdot R_m + 2 \cdot B = 241.921\ mm$$

$$l_{wmed1} := l_g + l_{f1} = 412.921\ mm$$

$$L_1 := 2 \cdot w_1 \cdot l_{wmed1} = 49.55\ m$$

$$R_1 := \rho_{115} \cdot \frac{L_1}{S_{Cu1} \cdot a_1} = 0.067\ \Omega$$

$$I_2 := I_b \qquad R_2 \cdot I_2^{\ 2} = R_b \cdot I_b^{\ 2} + 2 \cdot R_i \cdot I_i^{\ 2}$$

$$\rho_{115'} := 0.043125\ \Omega \cdot \frac{mm^2}{m}$$

$$\rho_{21} := 0.031\ \Omega \cdot \frac{mm^2}{m}$$

$$L_b := l_g = 171\ mm$$

$$h_{cr1} := h_{d1} = 28.4\ mm$$

$$h_{cr2} := h_{d2} = 19.05\ mm$$

$$B := -35\ mm \qquad p = 2$$

$$y_\tau := m_1 \cdot q_1 = 18$$

$$y_1 := \frac{5}{6} \cdot y_\tau = 15$$

$$S_{Cu1} = 9.079\ mm^2$$

$$h_{i.sc} := h_{d2} + 6.197\ mm = 25.247\ mm \qquad \boxed{h_{i.sc}} := 25.25 \cdot mm$$

$$b_{i.sc} := \frac{S_i}{h_{i.sc}} = 33.692\ mm \qquad \boxed{b_{i.sc}} := 33.7 \cdot mm$$

$$D_{i.sc} := D_r - h_{i.sc} = 248.65\ mm$$

$$L_i := \frac{\pi \cdot D_{i.sc}}{Z_2} = 10.278\ mm$$

$$R_i := \rho_{115'} \cdot \frac{L_i}{S_i} = \left(5.21 \cdot 10^{-7}\right)\ \Omega$$

$$R_b := \rho_{115'} \cdot \frac{L_b}{S_b} = \left(7.496 \cdot 10^{-5}\right)\ \Omega$$

$$R_2 := R_b + \frac{R_i}{2 \cdot \sin\left(\frac{\pi \cdot p}{Z_1}\right) \cdot \sin\left(\frac{\pi \cdot p}{Z_1}\right)} = \left(1.093 \cdot 10^{-4}\right)\ \Omega$$

$$b_{cr1m} := \sqrt{\frac{1}{2} \cdot \left({b_{cr1v}}^2 + {b_{cr1b}}^2\right)} = 8.189\ mm \qquad \boxed{b_{cr1m}} := 8.2 \cdot mm$$

$$h_{cr1v} := \frac{b_{cr1m} - b_{cr1v}}{b_{cr1b} - b_{cr1v}} \cdot \left(h_{cr1} - h_{istm1} - h_{pana} - 4 \cdot g_{iz}\right) = 12.18\ mm \qquad \boxed{h_{cr1v}} := 12.2 \cdot mm$$

$$h_{cr1b} := \frac{b_{cr1b} - b_{cr1m}}{b_{cr1b} - b_{cr1v}} \cdot \left(h_{cr1} - h_{istm1} - h_{pana} - 4 \cdot g_{iz}\right) = 10.52\ mm \qquad \boxed{h_{cr1b}} := 10.5 \cdot mm$$

$$K_{Cu} := 6 \cdot \frac{p \cdot y_1}{Z_1} + 1.67 = 4.17 \qquad K_h := 6 \cdot \frac{p \cdot y_1}{Z_1} + 1 = 3.5$$

$$\lambda_{cr1} := \frac{1}{4} \cdot \left(\frac{2}{3} \cdot \frac{h_{cr1b}}{b_{cr1b} + b_{cr1m}} + \frac{g_{iz}}{b_{cr1m}} + K_{Cu} \cdot \frac{h_{cr1v}}{b_{cr1m} + b_{cr1v}}\right) + \frac{1}{4} \cdot K_h \cdot \left(\frac{h_{pana} + 2 \cdot g_{iz}}{b_{cr1v}} + \frac{2 \cdot h_{pana}}{b_{cr1v} + b_{istm1}} + \frac{h_{istm1}}{b_{istm1}}\right)$$

$$\lambda_{cr1} = 2.287$$

$$l_g = 171\ mm$$

$$k'_\beta := \frac{y_1}{y_\tau} = 0.833$$

$$\tau = 216\ mm$$

$$\lambda_{d1} := \frac{t_1}{11.9} \cdot \frac{{k_{w1}}^2}{k_C \cdot \delta} = 1.109 \qquad l_{f1} = 241.921\ mm$$

$$\lambda_{f1} := 0.34 \cdot \frac{q_1}{l_g} \cdot \left(l_{f1} - 0.64 \cdot \tau\right) = 1.237$$

$$X_{\sigma 1} := 0.158 \cdot f_1 \cdot {w_1}^2 \cdot \frac{l_g}{p \cdot q_1} \cdot 10^{-4} \cdot \frac{\Omega\ s}{m} \cdot \left(\lambda_{cr1} + \lambda_{d1} + \lambda_{f1}\right) = 0.188\ \Omega$$

$$b_{cr2} := \frac{b_{cr2v} + b_{cr2b}}{2} = 5.45\ mm$$

$$\lambda_{cr2} := \left(\frac{h_{cr2} - h_{istm2}}{3 \cdot b_{cr2}} + \frac{h_{istm2}}{b_{istm2}}\right) = 1.771$$

$$\lambda_{f2} := \frac{2.3 \cdot D_{i.sc}}{4 \cdot Z_2 \cdot l_g \cdot \sin\left(\frac{\pi \cdot p}{Z_2}\right) \cdot \sin\left(\frac{\pi \cdot p}{Z_2}\right)} \cdot \log\left(\frac{4.7 \cdot D_{i.sc}}{h_{i.sc} + 2 \cdot b_{i.sc}}\right) = 1.776$$

$$\lambda_{d2} := \frac{t_2}{11.9} \cdot \frac{1}{k_C \cdot \delta} = 1.237$$

$$X_{\sigma 2} := 7.9 \cdot f_1 \cdot l_g \cdot 10^{-6} \cdot \frac{\Omega\ s}{m} \cdot \left(\lambda_{cr2} + \lambda_{d2} + \lambda_{f2}\right) = 0.000323\ \Omega$$

$$X_m := \frac{U_1 - I_\mu \cdot X_{\sigma 1}}{I_\mu} = 11.47\ \Omega$$

$$K := \frac{12 \cdot \left(w_1 \cdot k_{w1}\right)^2}{Z_2} = 481.112$$

$$R'_2 := K \cdot R_2 = 0.053\ \Omega$$

$$X'_{\sigma 2} := K \cdot X_{\sigma 2} = 0.155\ \Omega$$

7. CALCULUL PIERDERILOR ȘI RANDAMENTULUI MAȘINII DE INDUCȚIE

7.1. Pierderile principale in fier

$B_{j1} = 1.465\ T$ $\quad B_{j2} = 1.336\ T$

$p_{10} := 2.4\ \frac{W}{kg}$ $\quad y_{Fe} := 7800\ \frac{kg}{m^3}$ $\quad k_j := 1.4$

$D_{ej1} := D_e = 410\ mm$

$D_{ij1} := D + 2 \cdot h_{d1} = 331.8\ mm$

$D_{ej2} := D - 2 \cdot h_{d2} = 236.9\ mm$

$D_{ij2} := D_{ir} = 150\ mm$

$$p_{j1} := p_{10} \cdot \left(\frac{f_1}{50}\right)^2 \cdot B_{j1}^2 \qquad p_{j1} := 5.15\ \frac{W}{kg}$$

$$p_{j2} := p_{10} \cdot \left(\frac{f_1}{50}\right)^2 \cdot B_{j2}^2 \qquad p_{j2} := 4.28\ \frac{W}{kg}$$

$$m_{j1} := y_{Fe} \cdot \frac{\pi}{4} \cdot \left(D_{ej1}^2 - D_{ij1}^2\right) \cdot l_g \cdot k_{Fe} = 57.729\ kg$$

$$m_{j2} := y_{Fe} \cdot \frac{\pi}{4} \cdot \left(D_{ej2}^2 - D_{ij2}^2\right) \cdot l_g \cdot k_{Fe} = 33.46\ kg$$

$P_{j1} := k_j \cdot p_{j1} \cdot m_{j1} = 416.23\ W$

$P_{j2} := k_j \cdot p_{j2} \cdot m_{j2} = 200.491\ W$

7.1.2 Pierderile principale in dintii statorici si rotorici

$p_{d1} := p_{10} \cdot \left(\frac{f_1}{50}\right)^2 \cdot B_{j1}^{\ 2}$ $\quad p_{d1} := 5.15\ \frac{W}{kg}$ $\quad k_{j'} := 1.8$

$p_{d2} := p_{10} \cdot \left(\frac{f_1}{50}\right)^2 \cdot B_{j2}^{\ 2}$ $\quad p_{d2} := 4.28\ \frac{W}{kg}$

$m_{d1} := y_{Fe} \cdot h_{d1} \cdot b_{d1} \cdot Z_1 \cdot l_g \cdot k_{Fe} = 13.861\ kg$

$m_{d2} := y_{Fe} \cdot h_{d2} \cdot b_{d2} \cdot Z_2 \cdot l_g \cdot k_{Fe} = 9.264\ kg$

$P_{d1} := k_{j'} \cdot p_{d1} \cdot m_{d1} = 128.494\ W$

$P_{d2} := k_{j'} \cdot p_{d2} \cdot m_{d2} = 71.372\ W$

7.1.3 Pierderile principale in fier

$P_{Fepr1} := P_{j1} + P_{d1} = 544.723\ W$

$P_{Fepr2} := P_{j2} + P_{d2} = 271.863\ W$

$P_{Fepr} := P_{Fepr1} + P_{Fepr2} = 816.586\ W$

7.2 Pierderile suplimentare in fier la functionare in gol

7.2.1. Pierderile de suprafata

$Z_1 = 72 \qquad Z_2 = 76$

$\beta_{01} := 0.325 \qquad \beta_{02} := 0.17 \qquad \frac{b_{istm1}}{\delta} = 6 \qquad t_1 = 12\ mm$

$B_{01} := \beta_{01} \cdot k_C \cdot B_\delta = 0.327\ T \qquad \frac{b_{istm2}}{\delta} = 2.727 \qquad t_2 = 11.322\ mm$

$B_{02} := \beta_{02} \cdot k_C \cdot B_\delta = 0.171\ T \qquad n_1 = 1500\ min^{-1}$

$p_{supr1} := 0.95 \cdot \left(\frac{(Z_1 \cdot n_1)}{10000}\right)^{1.5} \cdot (t_1 \cdot B_{01})^2$ $\quad p_{supr1} := 519.179\ \frac{W}{m^2}$

$p_{supr2} := 0.95 \cdot \left(\frac{Z_2 \cdot n_1}{10000}\right)^{1.5} \cdot (t_2 \cdot B_{02})^2$ $\quad p_{supr2} := 137.062\ \frac{W}{m^2}$

$P_{supr1} := 2 \cdot p \cdot \left(\frac{t_1 - b_{istm1}}{t_1}\right) \cdot \tau \cdot l_g \cdot k_{Fe} \cdot p_{supr1} = 52.831\ W$

$$P_{supr2} := 2 \cdot p \cdot \left(\frac{t_2 - b_{istm2}}{t_2} \right) \cdot \tau \cdot l_g \cdot k_{Fe} \cdot p_{supr2} = 16.689\ W$$

7.2.2. Pierderile de pulsatie in dinti

$$B_{puls1} := \frac{\gamma_1 \cdot \delta}{2 \cdot t_1} \cdot B_{d1} = 0.128\ T$$

$Z_1 = 72 \qquad Z_2 = 76$

$n_1 = 1500\ min^{-1}$

$$B_{puls2} := \frac{\gamma_2 \cdot \delta}{2 \cdot t_2} \cdot B_{d2} = 0.04\ T$$

$$P_{puls1} := 0.1 \cdot \left(\frac{Z_1 \cdot n_1}{1000} \cdot B_{puls1} \right)^2 \cdot m_{d1} \qquad \boxed{P_{puls1}} := 264.887\ W$$

$m_{d1} = 13.861\ kg$

$m_{d2} = 9.264\ kg$

$$P_{puls2} := 0.1 \cdot \left(\frac{Z_2 \cdot n_1}{1000} \cdot B_{puls2} \right)^2 \cdot m_{d2} \qquad \boxed{P_{puls2}} := 19.26\ W$$

7.3. Pierderile suplimentare in fier la functionarea in sarcina

$$P_{sFe} := 0.005 \cdot P_N = 275\ W$$

7.4. Pierderile totale in fier

$$P_{Fe} := P_{Fepr} + P_{supr1} + P_{supr2} + P_{puls1} + P_{puls2} + P_{sFe} = 1445.253\ W$$

7.5. Pierderile electrice principale la functionarea in sarcina

$$P_{el1N} := m_1 \cdot R_1 \cdot {I_{1N}}^2 = 1916.405\ W$$

$$P_{el2N} := m_2 \cdot R_2 \cdot {I_2}^2 = 1285.195\ W$$

$$P_{elN} := P_{el1N} + P_{el2N} = 3201.6\ W$$

7.6. Pierderile mecanice prin frecare si ventilatie

$$P_{mec} := 26.4 \cdot p \cdot \tau^3 \cdot 10^{-3} \qquad \boxed{P_{mec}} := 532.102\ W$$

$\tau = 21.6\ cm$

$$P := P_{Fe} + P_{elN} + P_{mec} = 5178.956\ W$$

$$\boxed{\eta_N} := \frac{P_N}{P_N + P} = 0.914$$

$\varepsilon_{\eta N} := 0.44\%$

8. CALCULUL CARACTERISTICILOR MAȘINII DE INDUCȚIE CU PARAMETRI CONSTANȚI

$$\Omega_1 := \frac{2\,\pi \cdot f_1}{p} = 157.08\ \frac{rad}{s}$$

$P_N = 55000\ W$

$n_N = 1480\ min^{-1}$

$$s_N := \frac{n_1 - n_N}{n_1} = 0.013$$

$$C_1 := 1 + \frac{X_{\sigma 1}}{X_m} = 1.016$$

$$M_N := \frac{30 \cdot P_N}{\pi \cdot n_N} \qquad \boxed{M_N} := 354.75\ J$$

$$I'_{2N} := \frac{m_2 \cdot w_2 \cdot k_{w2}}{m_1 \cdot w_1 \cdot k_{w1}} \cdot I_2 = 90.095\ A$$

$$\boxed{s_N} := \frac{R'_2 \cdot I'_{2N}}{k_E \cdot U_1} = 0.021$$

$$M := \frac{m_1 \cdot p \cdot U_1^{\,2} \cdot \dfrac{R'_2}{s_N}}{2\,\pi \cdot f_1 \cdot \left[\left(R_1 + C_1 \cdot \dfrac{R'_2}{s_N}\right)^2 + \left(X_{\sigma 1} + C_1 \cdot X'_{\sigma 2}\right)^2\right]} = [369.717]\ N \cdot m$$

$$\boxed{P_{el2N}} := s_N \cdot M_N \cdot \Omega_1 = 1178.091\ W$$

$$P_M := M \cdot \frac{2 \cdot \pi \cdot n_N}{60}$$

$$\boxed{P_M} := 50600\ W$$

$$I_0 := I_\mu = 19.81\ A$$

$I_\mu = 19.81\ A$

$$I_{0r} := I_0 = 19.81\ A$$

$$P_{el0} := m_1 \cdot R_1 \cdot I_0^{\,2} = 79.162\ W$$

$$I_{0a} := \frac{(P_{Fe} - P_{sFe}) + P_{el0} + P_{mec}}{m_1 \cdot U_1} = 2.571\ A$$

$$cos\varphi_0 := \frac{I_{0a}}{I_0} = 0.13$$

$$\boxed{I_0} := \sqrt{I_{0a}^{\,2} + I_{0r}^{\,2}} = 19.977\ A$$

$$I'_{2a} := \frac{U_1 \cdot \left(R_1 + C_1 \cdot \frac{R'_2}{s_N}\right)}{\left(R_1 + C_1 \cdot \frac{R'_2}{s_N}\right)^2 + \left(X_{\sigma 1} + C_1 \cdot X'_{\sigma 2}\right)^2} = 87.463\ A$$

$$I'_{2r} := \frac{U_1 \cdot \left(X_{\sigma 1} + C_1 \cdot X'_{\sigma 2}\right)}{\left(R_1 + C_1 \cdot \frac{R'_2}{s_N}\right)^2 + \left(X_{\sigma 1} + C_1 \cdot X'_{\sigma 2}\right)^2} = 11.657\ A$$

$$I_{1N} := \sqrt{\left(I_{0a} + I'_{2a}\right)^2 + \left(I_\mu + I'_{2r}\right)^2} = 95.375\ A$$

$$cos\varphi_N := \frac{I_{0a} + I'_{2a}}{I_{1N}} = 0.944$$

$$\varepsilon_M := \frac{\left|M - M_N\right|}{M} \cdot 100 = [4.048]$$

Eroarea relativă a cuplului in procente